Explainable Quality Assessment and Conversational Assistance for Sri Lankan Medicinal Plants

***Abstract*—This paper presents two complementary components of a mobile decision-support system for Sri Lankan medicinal plants: an explainable cascaded vision pipeline for visual quality assessment and a retrieval-augmented conversational assistant for herbal knowledge. The quality-assessment component routes a leaf image to a species-specific condition head and, when the leaf is predicted healthy, evaluates maturity. All stages employ ImageNet-pretrained EfficientNetV2-B0 backbones with Grad-CAM explanations. Held-out test accuracies are 89.27% (condition) and 91.74% (maturity). The conversational assistant stores plant knowledge as naturally segmented chunks embedded with text-embedding-3-large in ChromaDB. A language-detection and query-rewrite step produces a standalone English search query; the top-five chunks are gated by cosine similarity before a persona-conditioned answer is generated. On a twelve-question bilingual pilot the correct plant was retrieved for eleven questions and safety notes were consistently surfaced. Together the components form an integrated mobile workflow that combines visual quality cues with reliable, refusals-aware textual guidance while remaining strictly non-clinical.**

***Index Terms*—medicinal plants, EfficientNetV2, Grad-CAM, cascaded classification, retrieval-augmented generation, Ayurvedic knowledge, explainable AI, mobile decision support**

I. Introduction

Medicinal plants remain central to primary healthcare and traditional Ayurvedic practice in Sri Lanka [1]. Visual quality assessment (disease, deficiency, damage, and maturity) and reliable textual knowledge about preparation and safety are therefore of practical importance to farmers, collectors, practitioners, and students. Existing agricultural computer-vision studies commonly treat crop disease as a separate classification problem [2], [3], while conversational access to Ayurvedic knowledge has been explored through ontology-based stores and pattern-matching chatbots that either demand specialized query languages or fail on paraphrased questions [4], [5], [6], [7].

This paper describes two integrated components of a mobile decision-support prototype. The first is a cascaded visual-quality workflow that selects a species-specific condition head and, when the leaf is predicted healthy, assesses maturity. Grad-CAM heatmaps and curated knowledge-base entries supply visual explanations and suitability guidance. The second is a Herbal Knowledge Assistant System (HKAS) that employs retrieval-augmented generation (RAG) with a confidence gate, bilingual (English/Sinhala) query handling, and persona-conditioned answers (general public, practitioner, student, researcher). Both components share a common backend service layer and plant databases; a cultivation-recommendation module is planned but lies outside the present scope.

The contribution is therefore a technically coherent, explainable decision-support route from leaf imagery and natural-language questions to species-aware visual quality cues and grounded textual guidance. No clinical diagnosis or medicinal-efficacy claim is made; all outputs are intended as visual and informational support only.

II. Literature Review

Reviewed Sri Lankan systems primarily identify medicinal-plant species from leaf imagery [8], [9], while broader agricultural studies commonly address crop disease as a separate classification problem [3]. Misidentification of Ayurvedic plants remains a practical concern that CNN-based methods have sought to mitigate [10], [11], [12]. This work extends that scope through a cascaded visual-quality workflow that selects a species-specific condition head and assesses maturity only for leaves predicted as healthy. The contribution is therefore not a medical diagnosis system; it is an explainable visual decision-support component for six locally relevant species: betel, *Centella asiatica*, holy basil, *Kalanchoe pinnata*, *Murraya koenigii*, and neem.

The reviewed studies also informed the cascade design. Mahendran et al. reported that EfficientNetB0 outperformed MobileNetV3 and ResNet50 for a Sri Lankan medicinal-plant task [8]. Jung et al. demonstrated a stepwise crop, disease-presence, and disease-type pipeline with an unknown category [3], while Yadav and Khadse combined an unknown class with Grad-CAM for medicinal-plant recognition [13]. The present work differs by connecting species-specific condition classification, maturity assessment, visual explanations, and knowledge-based suitability guidance within one mobile workflow.

Several studies have looked at giving Sri Lankan users conversational access to Ayurvedic knowledge. Ontology-based representations provide deep structured knowledge covering plants, plant parts, diseases and preparation methods, but users must write SPARQL queries, which is not practical for the general public [4], [5]. Sinhala chatbots and mobile assistive tools built with pattern matching or template responses match questions against pre-written patterns and answer with stored templates [6], [7]. This makes the systems easy to use, but they can only answer questions that match a pattern the developers have already written; a question phrased differently cannot be answered even if the needed information exists.

Retrieval-Augmented Generation (RAG) offers a different approach, where a language model generates an answer using passages retrieved from a knowledge base at the time of

the question, instead of matching fixed patterns [14], [15]. While language models understand natural language well, they can hallucinate information that is not present in any reliable source, which is a serious concern for medical-related applications [16]. The conversational assistant component uses a RAG-based approach so that paraphrased, or previously unseen questions can still be answered correctly, and includes a confidence check so the system declines to answer when nothing relevant is found in the knowledge base, rather than generating an unsupported answer.

III. Methodology

A. System Architecture

The mobile application captures a plant image or a text/voice question and routes it to the relevant component. The quality-assessment path and the conversational assistant both call into a common set of AI/backend services (model inference and API endpoints) and read from shared plant and image databases before returning a result to the mobile application. A cultivation-recommendation module is part of the same planned architecture but is not covered in this paper. The quality-assessment service accepts one to three leaf images, runs the cascaded EfficientNetV2-B0 pipeline, and returns stage-specific predictions, Grad-CAM visualizations, and knowledge-base entries. The conversational service accepts a natural-language question, performs language detection and query rewriting, retrieves relevant knowledge chunks, applies a confidence gate, and generates a persona-conditioned answer.

B. Cascaded Visual Quality Assessment

The API accepts one to three leaf images. Each image is letterboxed to 224×224 pixels, and probabilities from multiple views are averaged. Accepted samples are routed to one of six species-specific condition heads containing 36 healthy, disease, deficiency, discoloration, and damage labels in total. The configured acceptance threshold is 0.80. For a healthy condition, a subsequent stage uses five maturity heads (holy basil is presently disabled): four distinguish Young and Mature, while *Murraya koenigii* additionally includes Developing. Optional leaf texture and measured size are used only as rule-based maturity evidence, not as neural-network inputs.

All deployed stages use ImageNet-pretrained EfficientNetV2-B0 backbones with frozen feature extractors and softmax output heads [17]. Training used Adam at a learning rate of 0.001, with random flipping, rotation, zoom, and contrast augmentation. Batch sizes were 16 for condition and maturity; configured training limits were 15 epochs. Grad-CAM is generated from the selected condition class for diseased leaves or the maturity class for healthy leaves, highlighting influential image regions [18]. Disease and maturity records are then retrieved from curated CSV knowledge files. Any displayed medicinal-suitability value is rule-derived visual guidance and must not be interpreted as a clinical probability.

The final split folders contain 12,379 condition images (8,003/2,131/2,245 train/validation/test) and 848 maturity images (620/107/121). Most labels were inherited from source datasets and reorganized into stage-specific classes.

C. Conversational Assistant

The conversational assistant knowledge base stores one entry per medicinal plant, with each entry containing its description, medicinal uses, preparation method and precautions. Entries are split into separate chunks along their natural boundaries rather than by a fixed character length, so that all fields belonging to one plant stay together. Each chunk is converted into an embedding using OpenAI's text-embedding-3-large model and stored in a ChromaDB vector database.

When a user asks a question, the system first checks whether it is written in Sinhala or English. A single language-model call then rewrites the question into a standalone English search query, using the last two turns of the conversation to resolve references such as "what about for skin?", and translating the question if it was written in Sinhala. This rewritten query is used to retrieve the five most similar chunks from the vector database. If none of the retrieved chunks are similar enough (measured using a cosine distance threshold), the system returns a fixed message saying it does not have reliable information, instead of attempting to guess an answer. For Sinhala questions, the system also checks that at least one Sinhala word from the question appears in the retrieved text, as an extra safeguard against retrieving the wrong plant.

Once relevant chunks are retrieved, a second language-model call generates the final answer. The answer is generated according to one of four personas—general public, practitioner, student, or researcher—so that the level of detail matches the type of user. The prompt also instructs the model to stay on the plant identified in the question and to mention any precaution or safety information found in the retrieved text. The system keeps the last three question-answer pairs of each conversation in memory, so follow-up questions do not require the plant name to be repeated. Fig. 1 shows this process.

IV. Results and Discussion

Table I reports held-out results for the active EfficientNetV2-B0 checkpoints. Condition and maturity statistics are support-weighted summaries of their species-head reports rather than metrics from a single flat classifier.

TABLE I
Active EfficientNetV2-B0 Test Results

Stage	Test n	Accuracy	Macro F1
Condition*	2,245	89.27%	88.37%
Maturity*	121	91.74%	92.42%

*Support-weighted aggregation across species-specific heads.

MobileNetV3-Large produced slightly higher isolated condition metrics, but EfficientNetV2-B0 was retained for a consistent deployed cascade and achieved the strongest saved maturity result. A manually recorded 17-image field pilot

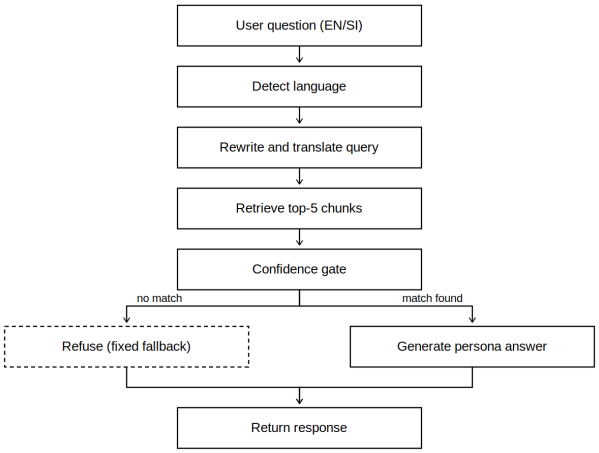
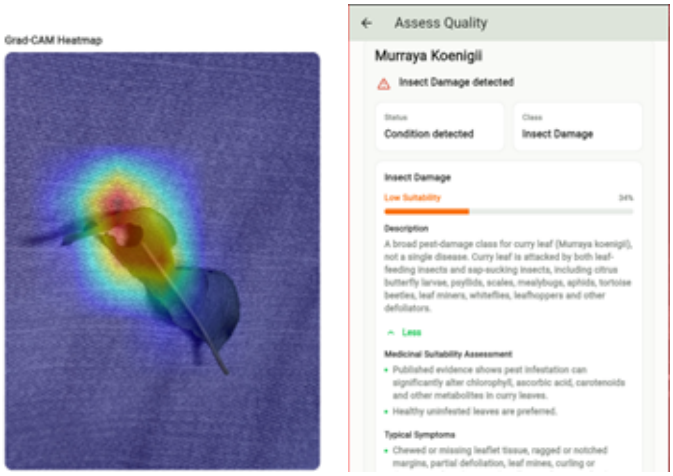


Fig. 1. conversational assistant query-processing pipeline: language detection, query rewrite/translation, top-5 retrieval, confidence gate, and persona-conditioned answer generation.



(a) Grad-CAM heatmap (b) Quality assessment result

Fig. 2. *Murraya koenigii* condition output: (a) Grad-CAM heatmap highlighting the damaged leaflet region; (b) corresponding mobile interface showing insect damage prediction with 93.4% condition confidence and a rule-derived 34% suitability score.

favored the all-EfficientNet cascade; because its sampling and execution artifact were not archived reproducibly, it is treated as exploratory evidence rather than a primary performance result.

The aggregated condition value conceals substantial variation among species heads. Test accuracy ranged from 76.84% for betel ($n = 190$) and 86.85% for neem ($n = 1{,}369$) to 100% for holy basil ($n = 135$). The maturity heads ranged from 84.62% for *Murraya koenigii* ($n = 39$) to 100% for betel ($n = 34$). These figures indicate that the condition stage is the main performance bottleneck and that the apparently strong maturity aggregate must be interpreted cautiously because each species has only 16–39 test samples.

For diseased leaves the system returns the condition label, instance confidence, Grad-CAM heatmap, typical symptoms, comparison with a healthy leaf, handling guidance, and treatment information. Healthy predictions expose the maturity decision and corresponding explanation. The application deliberately separates model confidence from the knowledge-base suitability indicator: for example, 93.4% is the neural condition confidence, whereas 34% is a deterministic suitability score. Neither value is overall model accuracy. Fig. 2 shows a *Murraya koenigii* sample predicted as insect damage together with its Grad-CAM explanation. The heatmap concentrates activation around the visibly damaged leaflet region, providing a qualitative check that the prediction was influenced by the plant rather than only by the background. This example demonstrates interface integration, not statistical validation of explanation quality. Formal expert scoring or localization measures were not collected.

The Flask service returns stage-specific predictions to a Flutter interface. Low-confidence or rejected predictions stop downstream assessment and request another view or expert verification. Diseased predictions return the condition label, instance confidence, Grad-CAM, typical symptoms, comparison with a healthy leaf, handling guidance, and treatment information. Healthy predictions expose the maturity decision and corresponding explanation.

The conversational assistant component was tested using six medicinal-plant questions, each asked in both English and Sinhala, giving twelve questions in total. The correct plant was retrieved and used to answer eleven of these questions, with no case where the system answered using information about the wrong plant. One Sinhala question was incorrectly refused by the confidence check even though the same question in English was answered correctly, suggesting the similarity threshold may need further tuning for some plants. Where the retrieved text contained a precaution or safety note, this was included in the generated answer in every relevant case. Fig. 3 shows a sample exchange.

The cascaded visual pipeline demonstrates that a single backbone family can support fine-grained condition assessment and maturity estimation while remaining deployable on a mobile-oriented backend. The condition stage remains the principal bottleneck. Grad-CAM provides qualitative transparency useful for interface-level trust, but quantitative localization studies and expert scoring are still required. The maturity test set contains only 121 images, and neither exact nor perceptual duplicate leakage across partitions has been ruled out. Complete source-licence records, annotation provenance, and Ayurvedic expert validation are also unavailable; accordingly, no clinical-safety or medicinal-efficacy claim is made. The maturity configuration and saved label order should be reconciled before retraining. A further integration defect was found: the current backend invokes maturity whenever the top condition class is Healthy, without checking the condition accepted flag. Consequently, a low-confidence Healthy prediction could still reach maturity; that gate must be corrected and retested before final evaluation.

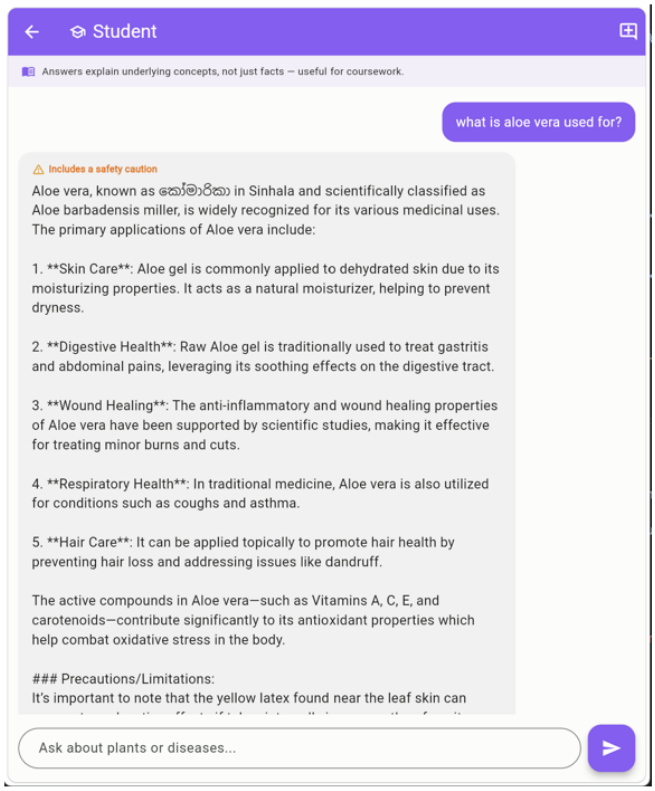


Fig. 3. Sample conversational assistant exchange (student persona) answering "what is aloe vera used for?" with structured uses and a safety caution.

Before the final paper is frozen, the dataset manifest should record every source URL, licence, original label, relabelling decision, and class mapping. SHA-256 and perceptual-hash audits should then identify cross-split duplicate groups. Final experiments should set Python, NumPy, and TensorFlow seeds; save exact package and GPU metadata; state whether augmentation occurred only after splitting; and archive the selected checkpoint with its matching configuration and label order. The corrected cascade should be re-evaluated on the held-out splits and on a preserved field set containing supported, unsupported, healthy, diseased, and low-confidence examples. Reporting confidence calibration and repeated-run variability would also distinguish reliable uncertainty handling from a single favorable training run. For the conversational assistant, the twelve-question pilot is insufficient for strong generalization claims, and the knowledge base itself has not yet undergone systematic expert review; a larger bilingual test suite with expert adjudication and calibrated similarity thresholds is required.

The conversational assistant results indicate that a RAG pipeline with an explicit confidence gate can answer both English and Sinhala questions while almost always retrieving the correct plant and surfacing safety information when present. Integration of the two components inside a shared mobile architecture is technically coherent: a user may capture a leaf, obtain an explainable quality assessment, and then ask follow-up questions about the same plant through the conversational interface.

V. Conclusion

The implemented prototype demonstrates a technically coherent route from leaf imagery to explainable visual-quality guidance and from natural-language questions to grounded, refusal-aware textual answers, with held-out accuracies of 89.27% and 91.74% for condition and maturity, respectively, and an eleven-out-of-twelve correct-plant retrieval rate on a bilingual pilot. Its defensible contribution is the cascaded, species-aware integration of visual quality assessment with retrieval-augmented knowledge assistance rather than a claim of clinical readiness. Reproducible field evaluation, leakage testing, provenance completion, gate correction, threshold calibration, and domain-expert validation remain necessary before deployment. In a pilot evaluation of 12 bilingual queries, the system retrieved the correct plant for 11 queries, yielding a preliminary top-1 retrieval rate of 91.7

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